MIL-C-915/62B
5 April 1973
SUPERSEDING[1]/
MIL-C-915/62A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE 3SF

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

(B)

First - Copper conductor, AWG No. 18 (7 X 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Nominal thickness 0.017 inch. Colored insulation - one conductor black, one conductor white and one conductor red for each triad.
Third - Insulation jacket of clear polyamide, type III grade E, or type IV of MIL-M-20693, nominal thickness 0.003 inch. Diameter over insulation jacket 0.086 inch nominal.
Fourth - One black, one white and one red conductor cabled together with a nominal lay of 1-1/2 inches to form a triad. Fillers of dry cotton as necessary in order to form a round assembly.
Fifth - One polyester tape, type G of MIL-I-631, applied helically with overlap.
Sixth - Braided shield of AWG No. 36 uncoated or tin-coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.004 inch. Standard Identification Code, applied by method 2.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard Identification Code applied by method 2 on inner tape.
- Diameter over shield insulation 0.240 inch nominal.
Eighth - The specified number of triads (see details) cabled together with a lay not greater than 16 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with triad No. 1, from center outward. A single triad shall not be used as the central core of the cable. Fillers of dry jute or dry cotton as necessary to form a firm, rounded assembly.
Ninth - Binder tape of polyester, type G of MIL-I-631, applied helically with overlap.
Tenth - Dry cotton braid, conforming to class A of Insulated Power Cable Engineers Association (IPCEA) Publication No. S-19-81.
Eleventh - Arctic type polychloroprene jacket (black) double layer, reinforced. Minimum thickness 0.100 inch. Cable surface marking required.

Details

Type and size	Number of triads	Overall diameter	
		Min. (inches)	Max. (inches)
3SF-7	7	1.000	1.040

[1]/ Requirements formerly covered on MIL-C-24145/22.

(B) Denotes changes.

FSC 6145

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EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft. at 25°C., max.	6.80
Voltage withstand - Volts, rms, minimum	
Conductor to conductor, within triads	2000
Conductor to shield, within triads	1000
Shield to shield, between triads	500
Group A:	
Visual and dimensional	
Mutual capacitance - Within triads, at 1 kHz, pF/ft., max.	30
Capacitance unbalance - Percent maximum	5
Group B:	
Cold bending, cable - At minus 30° ± 2°C., maximum mandrel diameter twelve times cable diameter	No damage
Bending endurance - At 25° ± 2°C., cycles, minimum	2000
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum	60
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTH: 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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